

Facial Emotion Detection Using DeepFace

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Abstract— Facial emotions are the most important signals of our face as they tell us about an individual's personality, feelings, motives, or intentions. Emotion detection offers a wide range of applications in human-computer interaction, healthcare, target advertising, education, and many more. This paper explores the use of DeepFace for facial emotion detection. DeepFace is a deep learning facial recognition and facial attribute analysis library which has pre-trained models. The performance is evaluated for static images, multi-face images, and in real-time integrated with user feedback to achieve better accuracy. DeepFace also identifies multi-faces in images and in real-time. We employ OpenCV and DeepFace analyzer for face detection and emotion detection. This model also gives the confidence scores in percentages of detected emotion labels. We achieved an accuracy of 84.62% for real-time with user feedback, 73.33% for static images and 55.55% for multi-face images.

Keywords— *facial emotion detection, DeepFace, OpenCV, static images, multi-face images, real-time, user feedback, DeepFace analyzer.*

I. INTRODUCTION

Emotions are the most complex and pivotal physiological responses that are generally strong feelings expressed by a person in response to a specific situation. Emotions can be expressed in many ways and their understanding is most important. In the field of computer vision and artificial intelligence this is becoming an important topic of study and area of research. Various machine learning models, including deep learning approaches are being used to detect emotions. DeepFace, a powerful open-source library, is an efficient and effective solution for facial recognition and emotion detection. It is introduced by Facebook's artificial intelligence research team in early 2015. DeepFace model has a 9-layer neural network and was trained on a very large dataset. The default DeepFace configuration uses VGG-Face model. Apart from emotion detection DeepFace can also be integrated to identify gender, age, and other facial attributes. Some of the earlier models have limitations with multi-face recognition, handling diverse lightning conditions, indicating confidence percentages and achieving high accuracy and real-time performance. The main goal is to address these issues by using this robust pre-trained DeepFace model which is capable of detecting emotions in static images, multi-face images and in real-time with user feedback. The emotions identified are-

happy, sad, angry, neutral, fear and surprise. The output shows the labels of detected emotion along with its confidence percentage. Integrating user feedback is an advantage for real-time emotion detection. Feedback from users on the correctness of predicted emotions helps in fine-tuning the model for better real-time performance. This aspect helps enhancing the system's ability to adapt and improve the accuracy overtime.

II. LITERATURE SURVEY

In [1] it mainly focused on facial emotion recognition in both real-time and static images where a hybrid approach was proposed which combines feature extraction using machine learning techniques. SVM based classification algorithm achieved high accuracy. The findings in [1] also suggest that fully-automatic user-independent real-time facial expression coding is achievable objective in continuous video stream with given computer capabilities. Use of Convolution Neural Network for expression recognition was explored in [2] showing how the deep learning can improve the detection. It did not include any preprocessing, feature extraction steps and used a six-layer CNN and got the accuracy of 61.7% with FER2013. In [3] to improve the accuracy of emotion recognition through facial expressions CNN is used. It uses iCV MEFED dataset and detected seven emotions by training and validating the dataset showing improvements in accuracy compared to traditional methods. The [3] findings state that CNN based approach can extract important features from facial expressions further improving accuracy. A system was proposed by [4] using DeepFace for live detection of emotions employing a webcam and also used several image processing techniques. The findings suggest high accuracy is achieved and it is also able to adapt varying lighting conditions and facial orientations. It uses a pre-trained model. In [5] the authors introduced a EmotionTracker, which is a real-time emotion detection system using OpenCV and DeepFace. They focused on improving accuracy and correctly identify emotions from live video feeds. They achieved significant accuracy and was able to track the emotions effectively. In [6] a method is presented which is based on deep learning that uses CNN for real-time emotion recognition. This paper aimed to improve the accuracy and speed of emotion detection with CNN feature extraction

capability and also included training a CNN on diverse dataset and improved the accuracy of results. In [7] they have examined various algorithms for facial emotion recognition in real-time and also compared traditional methods with deep learning approaches, showcasing the strengths and weakness of each of them. It involved testing of various algorithms on standard datasets to evaluate their performance in terms of accuracy, processing speed and computational requirements. Their findings stated that deep learning models performed well compared to traditional methods but also required additional resources for computation purpose.

III. METHODOLOGY

A. Workflow of facial emotion detection for Static and Multi-face images

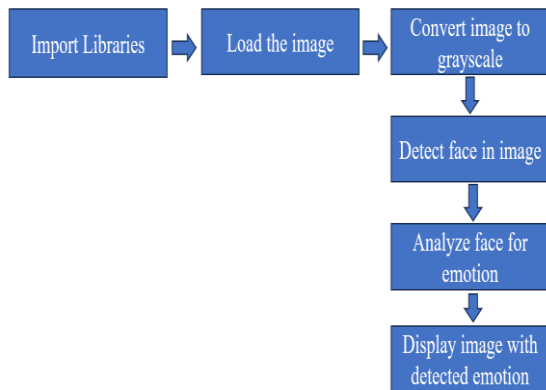


Fig. 1. Workflow of static & multi-face image emotion detection

This paper presents facial emotion detection using DeepFace for both static and multi-face images, where the user gives the path of the image. “Fig.1” shows the workflow of static and multi-face image detection where DeepFace, OpenCV, Matplotlib libraries are imported for this purpose. OpenCV is used for many purposes, one among them is to read the image path and convert the BGR image into gray if necessary. With the use of only one-color channel (gray) instead of three-color channel (BGR) simplifies the data. Resizing the images ensures uniform dimensions for all the input images which is essential for model to analyze them effectively. These preprocessing steps tend to reduce complexity and decrease the amount of data for the model to process, which results in faster computation and reduced memory usage. DeepFace uses OpenCV as the default face detector. Emotions for the detected faces are analyzed using `deepface.analyze`. As DeepFace can analyze other facial attributes and gender we indicate the action as [“emotion”] to meet specific objective. The function returns a list of results that include the detected emotions. A bounding box is drawn using OpenCV’s `CV2.rectangle`, which highlights the region of interest. The dimensions of the bounding box can be changed as per requirement. OpenCV is also used to show the labels of detected emotions along with the confidence score as a percentage. We can modify the font, color, and thickness as needed. The BGR image is converted to RGB after processing as OpenCV reads image as BGR which is the default color format, but Matplotlib reads images in RGB format, and that image is plotted at output. As the emotions are shown above the bounding box with their labels and

confidence percentages this output attained will provide a clear and simple view of the emotions that were detected. Multi-face image facial emotion detection includes the same processing steps, as in “Fig.1” but it returns a list of predictions where each entry corresponds to one detected face. Each prediction includes the coordinates of the face and its dominant emotion. The code will iterate through the list, draw bounding boxes, on each individual face and give labels for emotions detected.

B. Workflow of real-time facial emotion detection

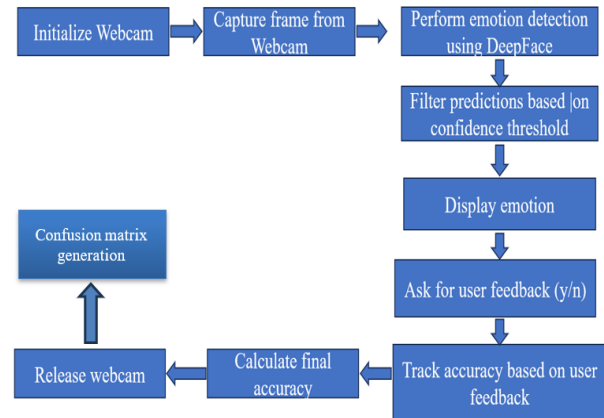


Fig. 2. Workflow of real-time emotion detection

“Fig.2” shows workflow for Real-time facial emotion detection using DeepFace which provides a thorough and interactive way to understand and comprehend the emotions captured through a webcam. This system uses DeepFace and OpenCV for emotion recognition and also for video capture and processing. The process begins with initializing the webcam, allowing continuous capture of frames, which are analyzed using the `DeepFace.analyze()` function. A confidence threshold is set to ensure that only reliable predictions are considered, while real-time feedback is provided and drawing bounding boxes around detected faces and displaying emotions on the screen. User feedback is an essential component of the system, allowing users to confirm or correct detected emotions by pressing ‘y’ or ‘n’ during the session. Correct and incorrect predictions are recorded, and the system tracks both overall and user-provided accuracy metrics. In this way, the system improves dynamically based on real-time user input. At the end of each session, a confusion matrix is generated that visualizes the model’s performance by comparing actual emotions (from user feedback) against predicted ones. In addition to this matrix the classification report, offers insights into the detection performance across a range of emotions like happy, sad, angry, fear, neutral and surprise. This method makes it possible to continuously monitor and enhance emotion detection making it useful tool for applications that need user inputs. With the integration of the confusion matrix and detailed feedback analysis, the system offers a comprehensive understanding of how well emotions are detected and aligned with actual human expressions.

IV. RESULTS

A. Image sources for static and multi-face image facial emotion detection

The static and multi-face images used in this work are sourced from Google. The images were gathered from publicly available platforms like Adobe Stock, iStock, Shutterstock, Freepik, VirtualFFS, BigStock, Westend61, Dreamstime, 123RF, PngTree for static images. Images showcasing Ekmanian facial expressions were sourced from Young et al.(2002) through ResearchGate as well as an image from KDEF dataset for multi-face static images. All images are mainly used in this work are for education purpose.

B. Output of Facial emotion detection for static images

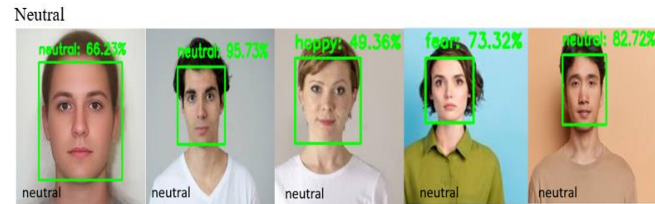


Fig. 3. Detections for neutral facial emotion

“Fig.3” shows 3 images were correctly identified as neutral and 2 wrong predictions were made.



Fig. 4. Detections for surprise facial emotion

In “Fig.4” the 5 predictions made as surprise were correct predictions.

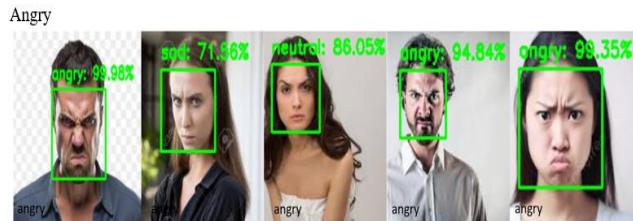


Fig. 5. Detections for angry facial emotion

The above “Fig.5” shows angry face images where 3 were correctly predicted as angry out of the 5 images given by user.



Fig. 6. Detections for sad facial emotion

“Fig.6” shows the sad facial emotion and 3 correct predictions were made as sad.

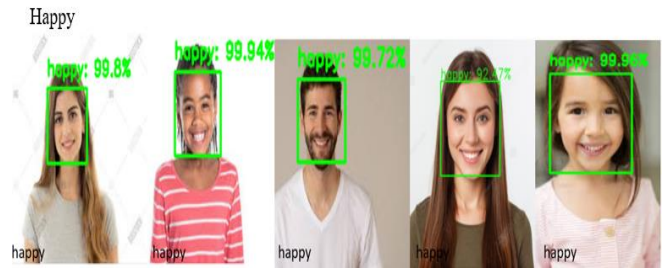


Fig. 7. Detections for happy facial emotion

As shown in “Fig.7” all the 5 predictions made as happy were correct.

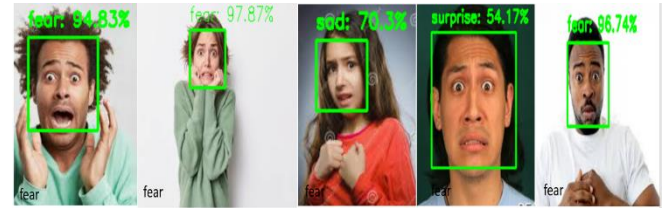


Fig. 8. Detections for fear facial emotion

In “Fig.8” only 3 images were correctly identified as fear out of the 5 images given.

C. Output of Facial emotion detection for multi-face images

Image 1-



Fig. 9. Multi-face image1 detections

The above “Fig.9” is a multi-face image with multiple emotions detected. Out of 6 emotions 2 emotions neutral and sad were incorrectly identified.

Image 2-

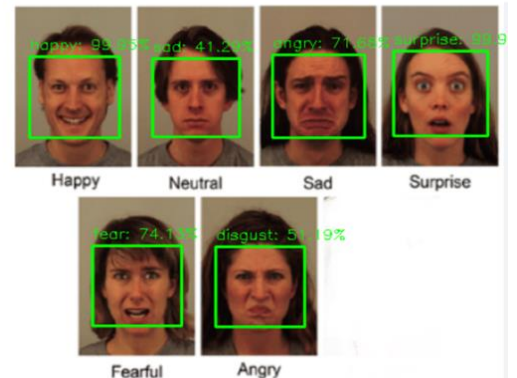


Fig. 10. Multi-face image2 detections

The “Fig.10” shows multi-face image2. Where 3 incorrect detections were made for neutral, sad, and angry. Image 3-

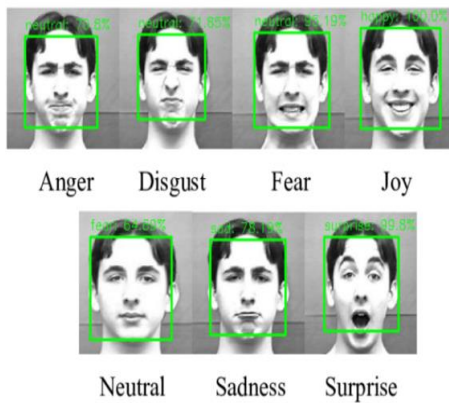


Fig. 11. Multi-face image3 detections

“Fig.11” shows multi-face image3. Here 3 incorrect identifications were made for anger, neutral, fear.

D. Output of Facial emotion detection in real-time

The below “Fig.12” shows the emotions detected in real-time through a webcam. The detected emotion labels were shown along with confidence percentage for 6 emotions-happy, sad, angry, neutral, surprise, and fear.

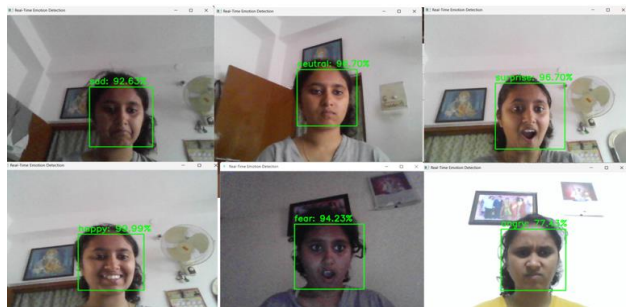


Fig. 12. Real-time emotion detection

“Fig.13” show the multi-faces identified in real-time through webcam where detected emotions labels and confidence percentages are displayed.

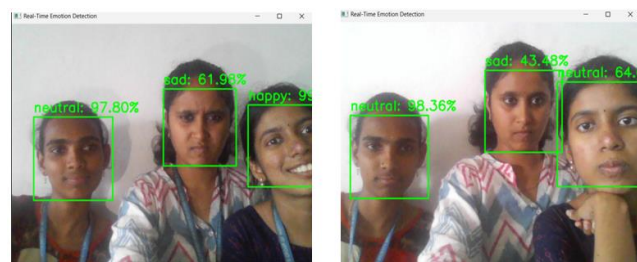


Fig. 13. Multi-face emotion detection in real-time

E. Accuracy of static image facial emotion detection

A set of 5 images are given for each emotion where the path of image is provided by the user. The findings show that the model misidentified between emotions like surprise, fear and sad and neutral. Happy and surprise emotions are accurately detected with highest confidence score and no wrong predictions are made. Table I shows the confidence percentages of detected emotions. Overall, it identified six

emotions: happy, sad, angry, neutral, surprise, and fear. Overall accuracy is calculated using the below equation-

$$\text{overall accuracy} = \frac{\text{total number of correct predictions}}{\text{total number of predictions}} * 100 \quad (1)$$

Here the model totally made 30 detections and 22 were correct detections and the total accuracy calculated for static images is 73.33% using “(1)”.

TABLE I. CONFIDENCE PERCENTAGES OF STATIC IMAGES

Emotions	Image 1	Image 2	Image 3	Image 4	Image 5	Highest %
Happy	99.80	99.94	99.72	92.47	99.96	99.96
Sad	99.49	65.12	78.01	Neutral (97.69)	Fear (95.24)	99.49
Angry	99.98	Sad (71.56)	Neutral (86.05)	94.84	99.35	99.98
Surprise	100	97.09	100	96.57	99.99	100
Fear	94.83	97.87	Sad (70.3)	Surpris e (54.17)	96.74	97.87
Neutral	66.23	95.73	Happy (49.36)	Neutral (82.72)	Fear (73.32)	95.73

F. Accuracy of multi-face image facial emotion detection

For the purpose of facial emotion detection in multi-face images three images with multi-faces are given by user with image paths. Each image has 6 faces within it. It has identified happy, sad, angry, fear, surprise and neutral. The model identified surprise and happy effectively with highest correct predictions. Table II shows the predictions made by the model . It struggles to identify sad, angry and neutral emotions meanwhile achieved slightly better for fear emotion. Accuracy is calculated using the equation-

$$\text{overall accuracy} = \frac{\text{total number of correct predictions}}{\text{total number of predictions}} * 100 \quad (2)$$

Using “(2)” we calculated accuracy where we achieved 55.55% for multi-face emotion detection. Here, total predictions made are 18 and correct predictions were 10.

TABLE II. MULTI-FACE STATIC IMAGES DETECTIONS

Emotion	Correct Detections	Total instances	Accuracy (%)
Happy	3	3	100.0
Sad	1	3	33.33
Angry	1	3	33.33
Surprise	3	3	100.0
Fear	2	3	66.67
Neutral	0	3	0

G. Accuracy of real-time facial emotion

Real-time detection outperforms multi-face detection, and even slightly surpasses static images in accuracy as it uses user feedback. This may be attributed to real-time feedback and dynamic context that enhances recognition. The confusion matrix accuracy is found using the equation –

$$cm_accuracy = \frac{np.trace(cm)}{np.sum(cm)} * 100 \quad (3)$$

In this calculation, using equation “(3)” where $np.trace(cm)$ computes the sum of matrix’s diagonal values which represent correctly classified emotions. This measures the accuracy by focusing on true positives across emotion classes. $np.sum(cm)$ calculates the total predictions by adding all values in the confusion matrix thus encompassing both accurate and inaccurate predictions. Together it provides the overall accuracy based on true positive classifications relative to total predictions. User feedback accuracy is found by using the equation –

$$user_feedback_accuracy = \frac{correct\ user\ feedback}{total\ user\ feedback} * 100 \quad (4)$$

Equations “(3)” and “(4)” were included in the code which gives the calculated accuracies at output. “Fig.14” shows the confusion matrix at output. The accuracy may further more increase based on user feedback. Multi-faces are also detected in real-time where each face detected is considered as individual face. “Fig.15” the classification report which shows accuracy of 84.62% through user feedback given by system and the confusion matrix accuracy is 84.62%.

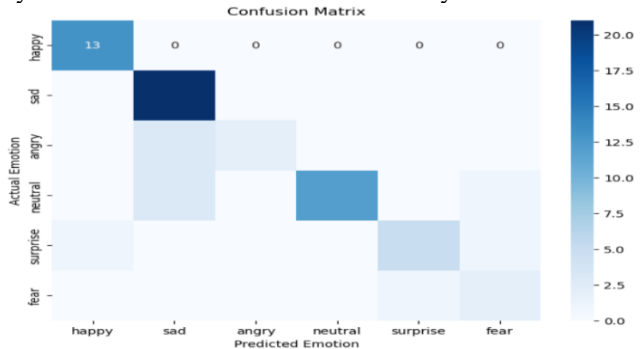


Fig. 14. Confusion matrix of real-time detected emotions

Classification Report:				
	precision	recall	f1-score	support
happy	1.00	0.40	0.57	5
sad	0.50	0.67	0.57	3
angry	0.93	1.00	0.96	13
neutral	1.00	0.75	0.86	16
surprise	0.78	1.00	0.88	21
fear	0.83	0.71	0.77	7
accuracy			0.85	65
macro avg	0.84	0.76	0.77	65
weighted avg	0.87	0.85	0.84	65

Confusion Matrix Accuracy: 84.62%
User Feedback Accuracy: 84.62%

Fig. 15. Classification report of real-time emotions detected

V. CONCLUSION

Facial Emotion Detection using DeepFace demonstrates a significant improvement in detecting and analyzing human emotions through facial expressions. By leveraging the DeepFace model, emotions in static images, multi-face images and real-time with use of webcam were successfully detected achieving an overall accuracy of 73.33% for static images and 55.55% for multi-face images and 84.62% for real-time detection with user feedback and multiple faces were also detected in real-time. With the implementation of a confusion matrix and classification report, the system provides a more detailed evaluation of prediction performance, offering insights into accuracy, precision, recall, and overall reliability. The use of confidence thresholds and real-time adjustments enhances the model's effectiveness, making it adaptable to different users and improving the detection emotions. These results highlight the model's capability to effectively analyze and classify emotions. Accuracy can vary significantly based on user interactions, emphasizing the importance of continuous learning in emotion detection systems and also it identifies areas for improvement. Notably, the performance varies across different emotions, with certain emotions like happy and surprise achieving near-perfect accuracy, while others, such as sad and neutral, displayed lower accuracy rates. This variance underscores the need for further enhancements in emotion classification, particularly for emotions that are often misidentified.

VI. FUTURE SCOPE

The Facial Emotion Detection using DeepFace has potential for improvements. One important area is enhancing of emotion detection capabilities, allowing for the detection of subtle emotional states and mixed emotions, which would provide a better understanding of human emotions. Using larger and more diverse datasets will improve the model performance. The identification of multi-face images can be much improved. Future work will also explore the integration of facial emotion detection with other models such as voice analysis and physiological signals, and YouTube video play according to detected emotion. Additionally, enhancing the existing user feedback mechanism by incorporating machine learning techniques that adapt the model based on user corrections could make the system more responsive to individual users' emotional recognition styles. As the technology advances, addressing ethical considerations, particularly regarding user privacy and consent, will be

imperative and should focus improving the accuracy of overall system.

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